Time-series Analytics

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Introduction

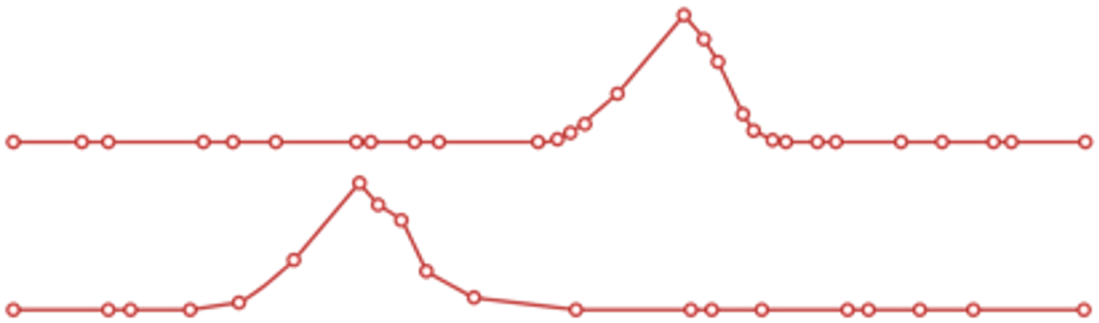
Type of data

A ***time series*** is a sequence of observations on **one** (or more) **quantitative** variable **regularly** collected **over time**.

Events vs time series

The phenomenon happens and we observe them

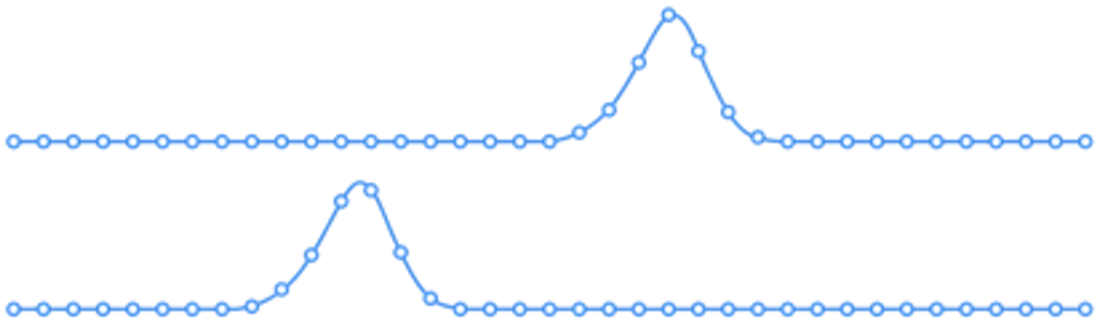
irregularly



Events

We monitor a phenomenon

regularly

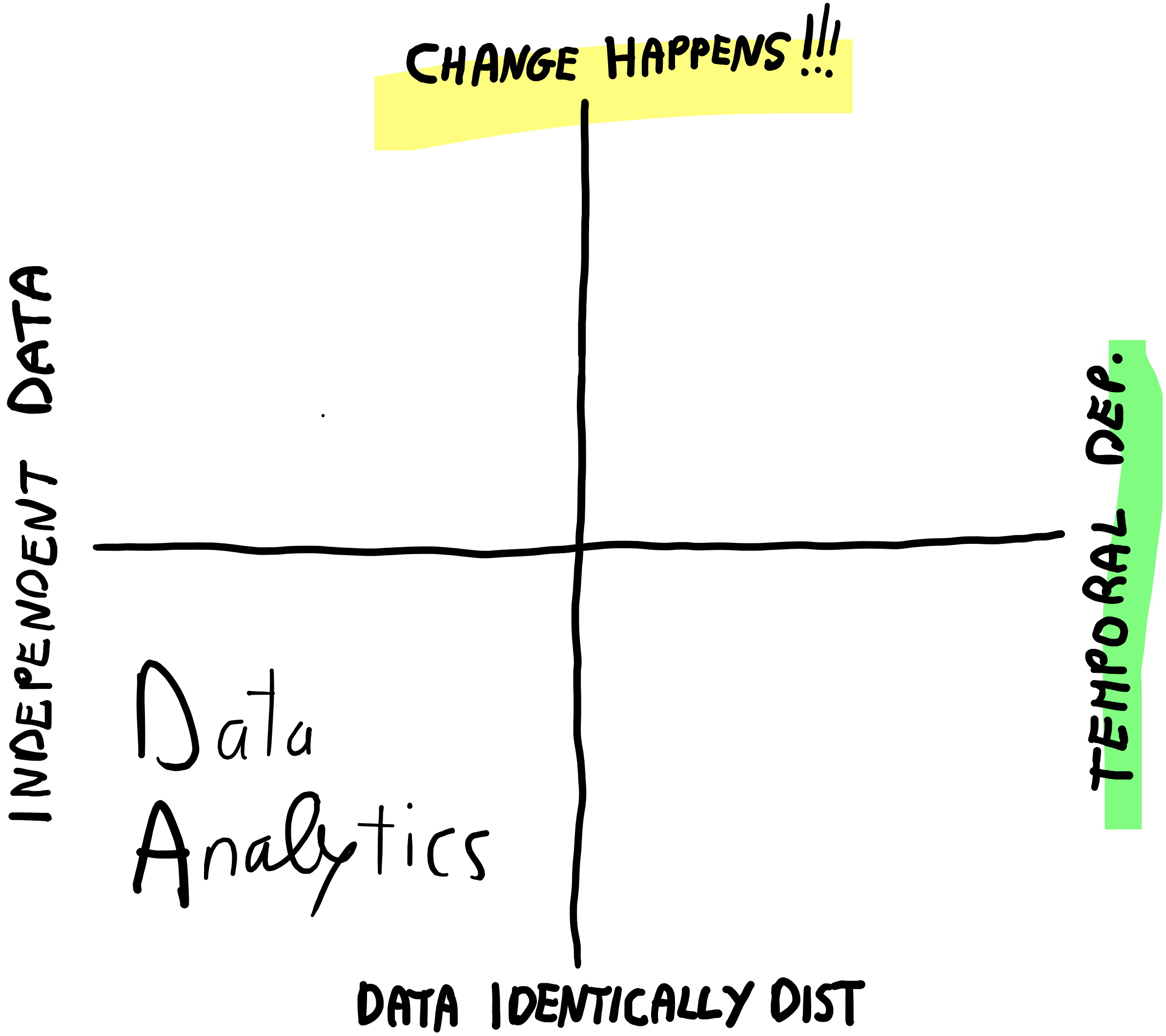


Time series

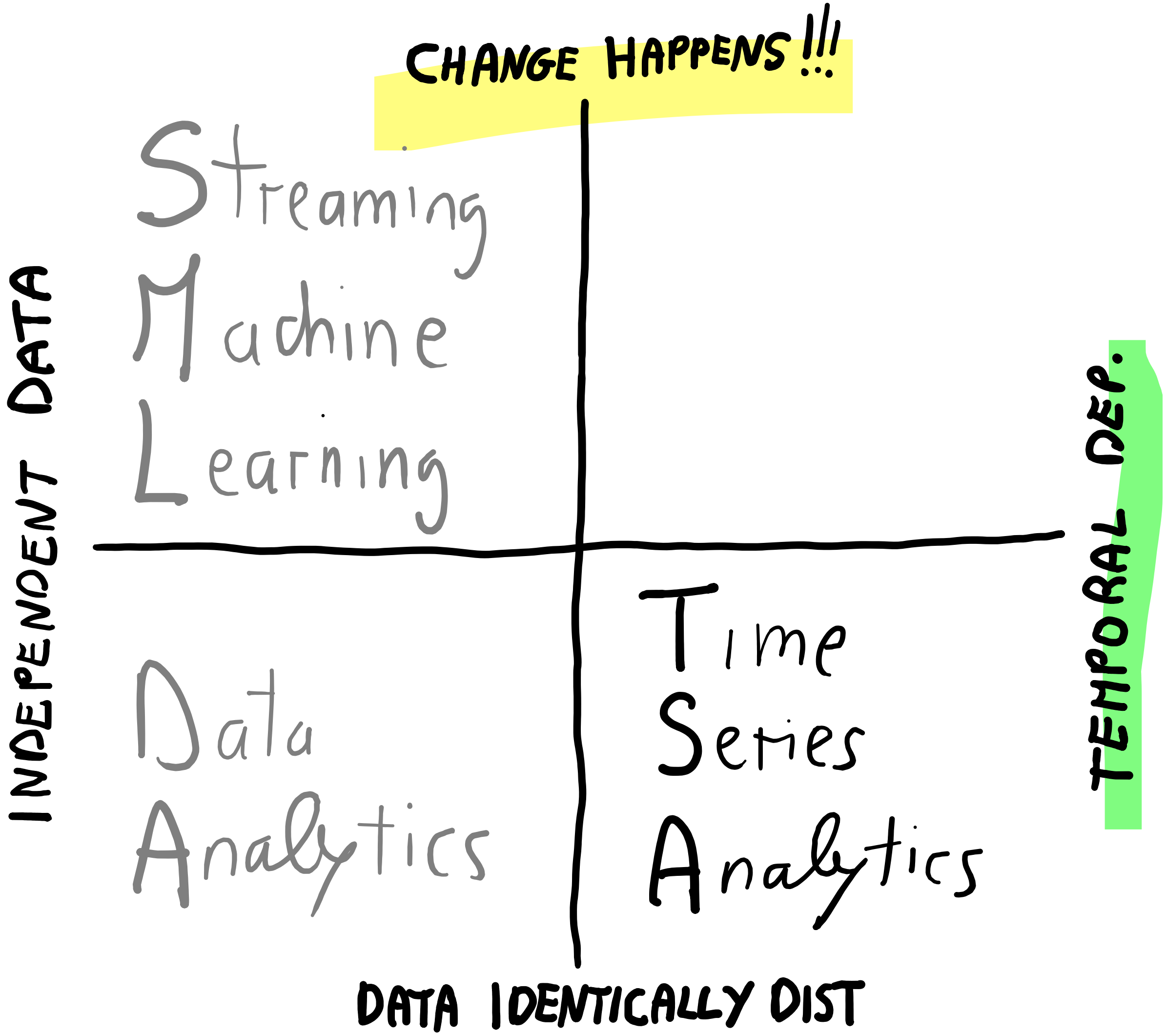
Time Series (brief recall from early lectures)

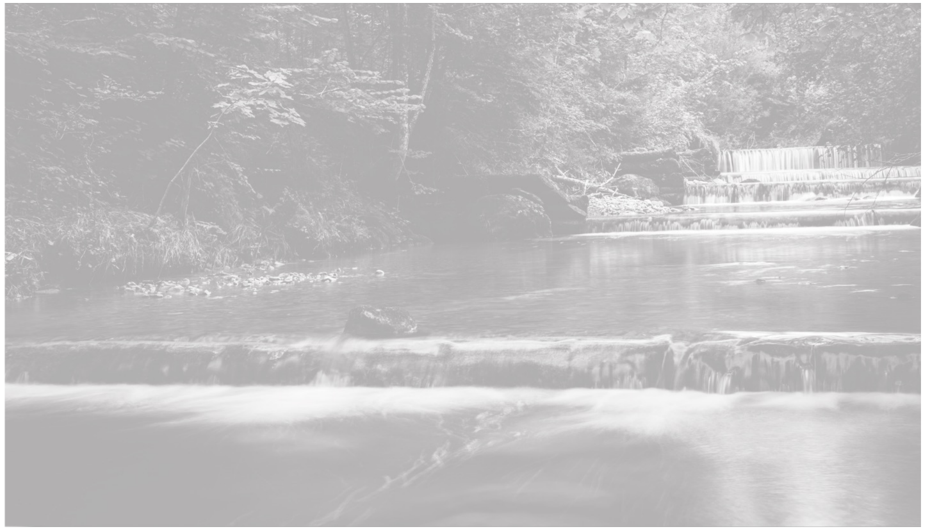
- A *time-series* is a set of observations on a quantitative variable collected over time.
- Examples
  - Dow Jones Industrial Averages
  - Historical data on sales, inventory, customer counts, interest rates, costs, etc.
  - Signal processing, pattern recognition, econometrics, mathematical finance, weather forecasting, earthquake prediction, electroencephalography, communications engineering, …
- Businesses are often very interested in analyzing and forecasting time series variables

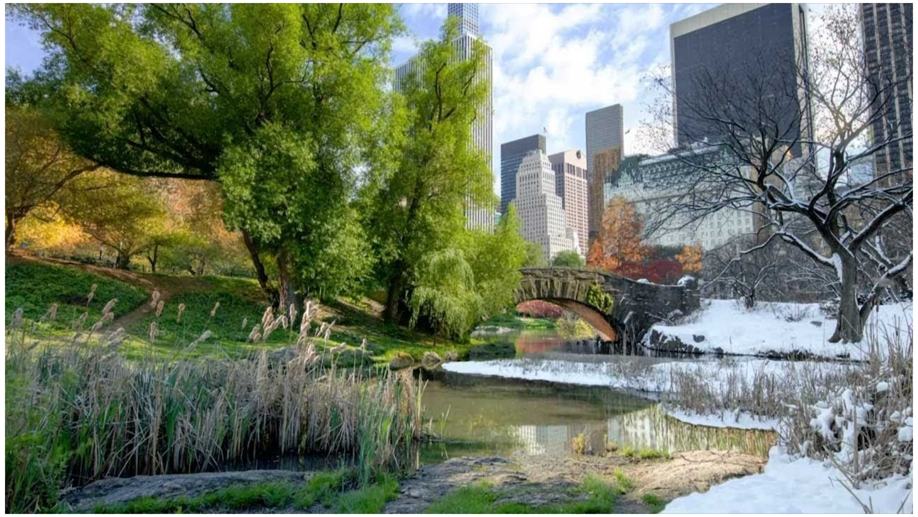
State-of-the-art



State-of-the-art







Explaining the past and forecast the future
of a continuous flow of data
without assuming data independence
with
Time Series Analytics

1st foundational concept

Stationarity

Fact

If a time series is stationary,… *… it is predictable*

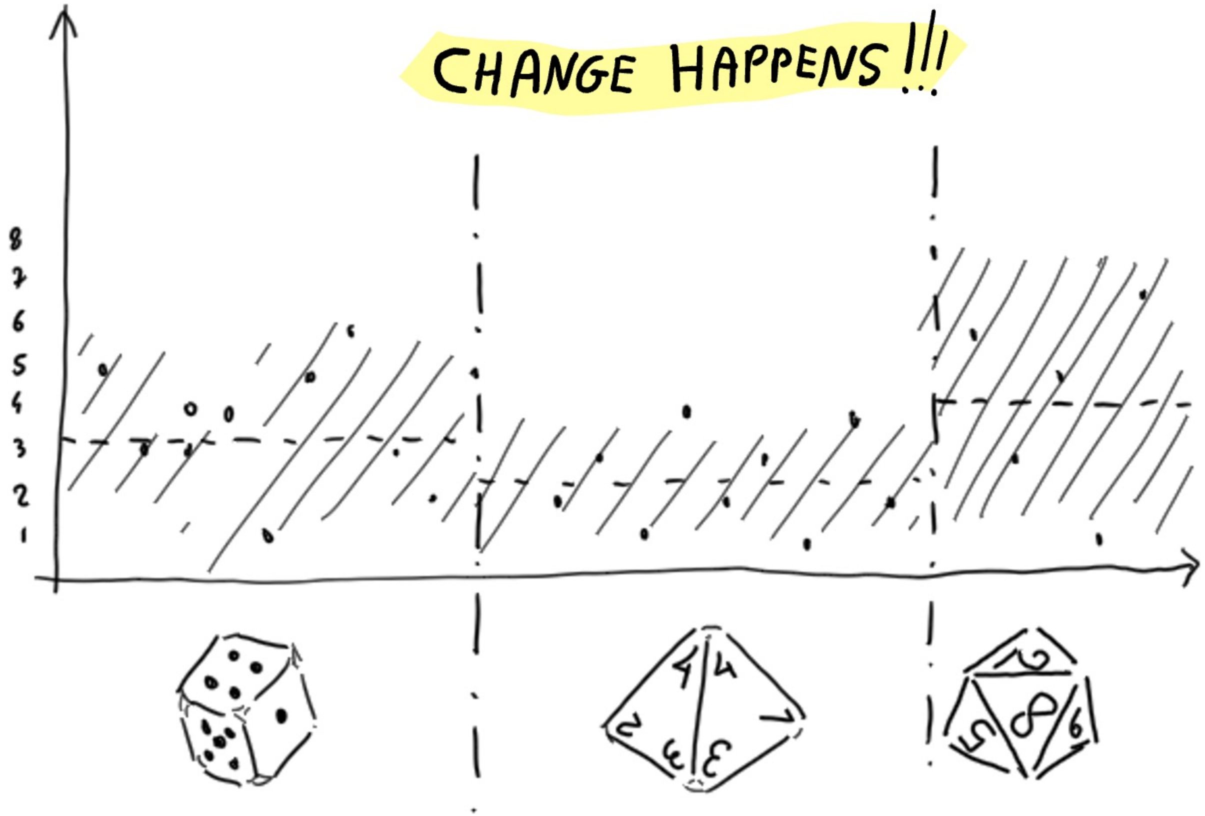
Photo by **cottonbro** from **Pexels**

Photo by **Bogdan Dirică** from **Pexels**

Definition

A **stationary** time series is one whose **properties do not depend on the time** at which the series is observed.

From identical distribution to stationarity



Assuming identical distribution means all observations come from the same distribution, no change across samples.

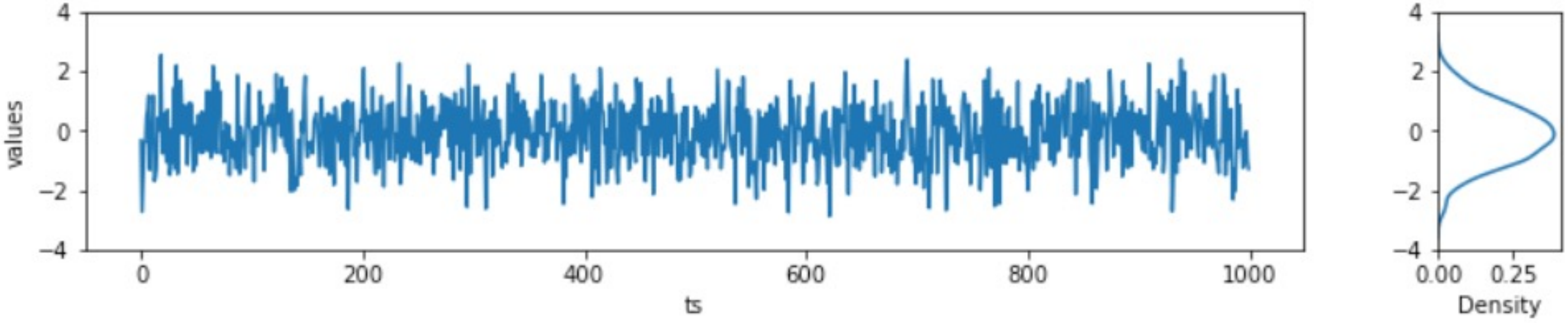
In a stationary time series, the **distribution doesn't change over time.**

- (i.) i.d. → identical distribution **across samples**
- Stationary process → identical distribution **across time**

Let’s build the intuition of **stationarity**

Stationarity
White noise: the perfect time series

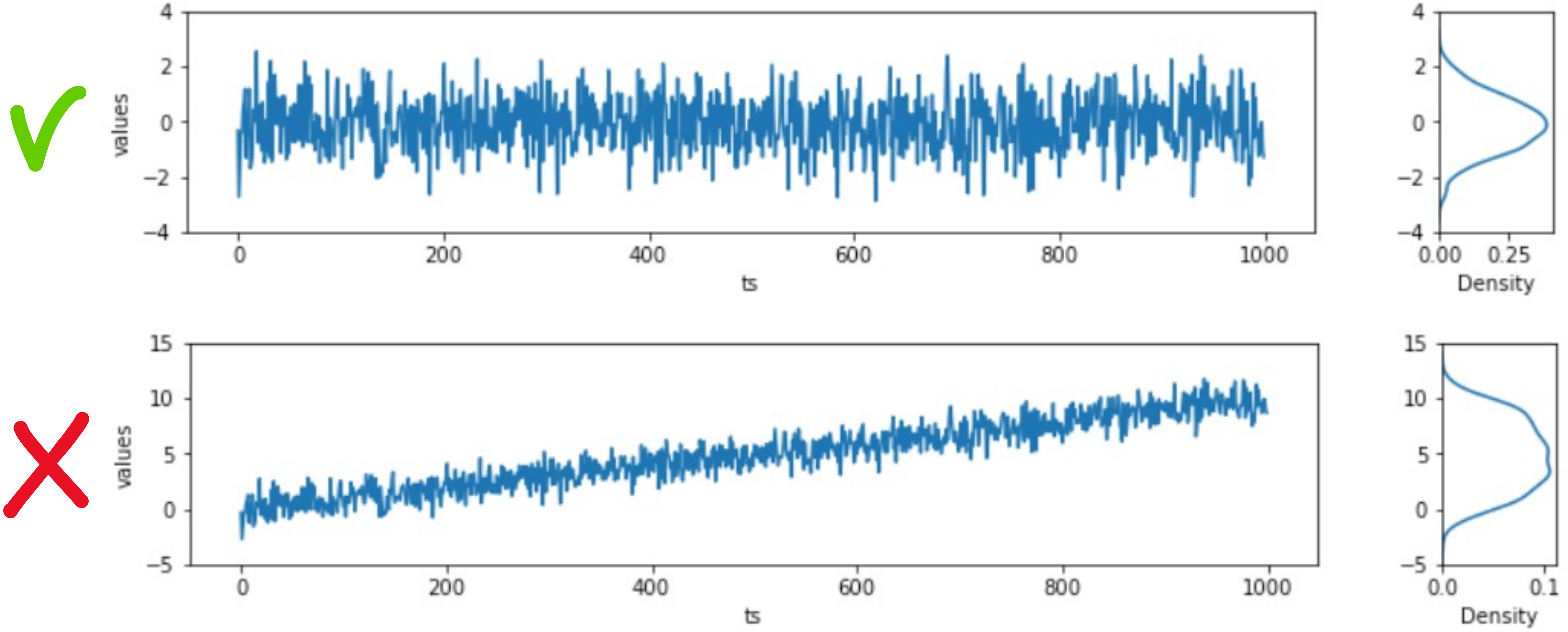
- A sequence of random numbers with zero mean and finite variance



- NOTE: it is a perfect example of a stationary time series
  - If you predict 0 (the mean), you minimize the error (which is proportional to the variance)

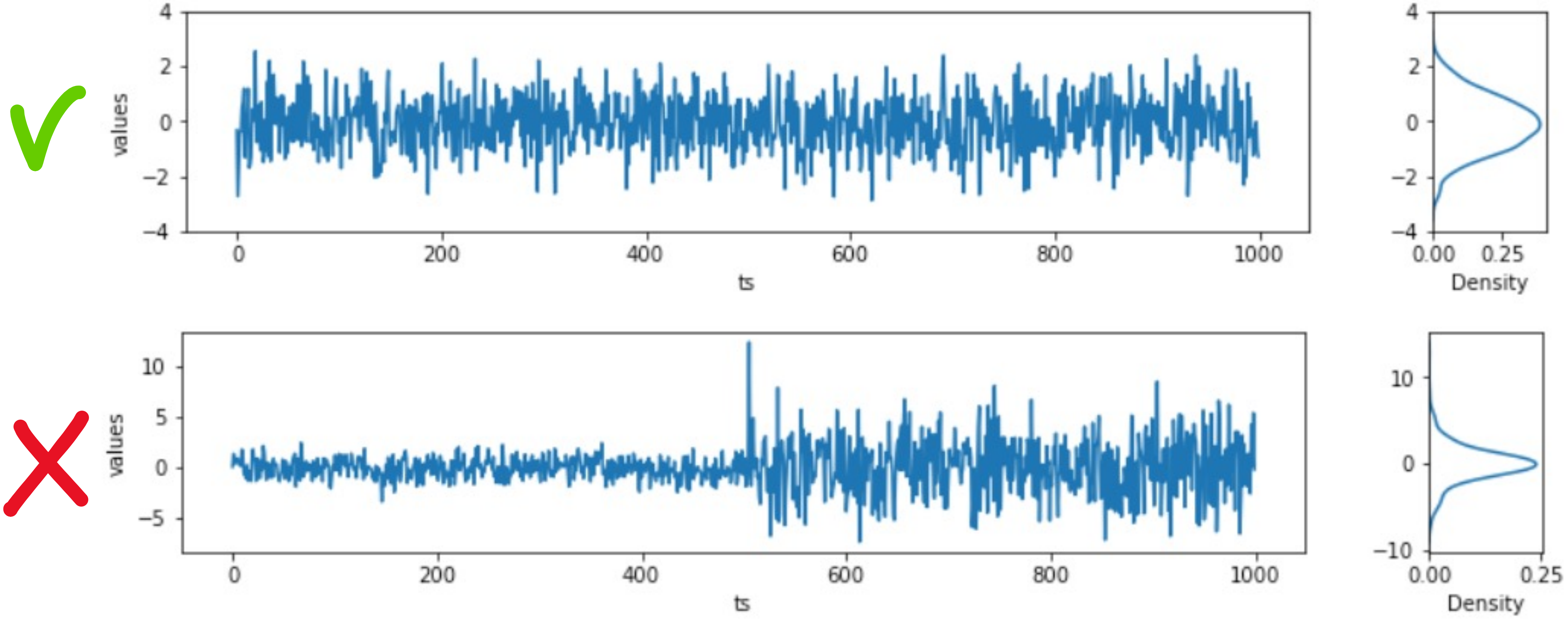
Stationarity

Mean is **constant** over time (a.k.a., with trend)



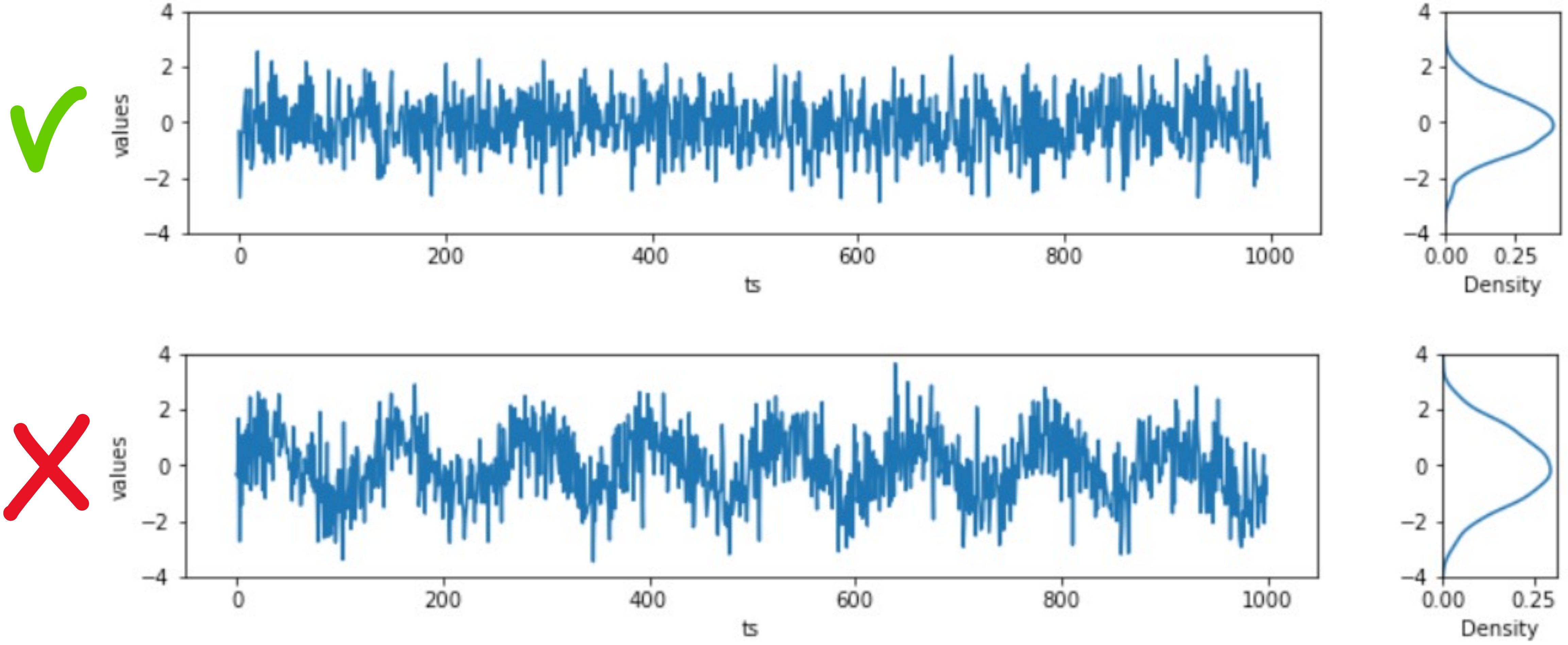
Stationarity

Variance is **constant** over time



Stationarity

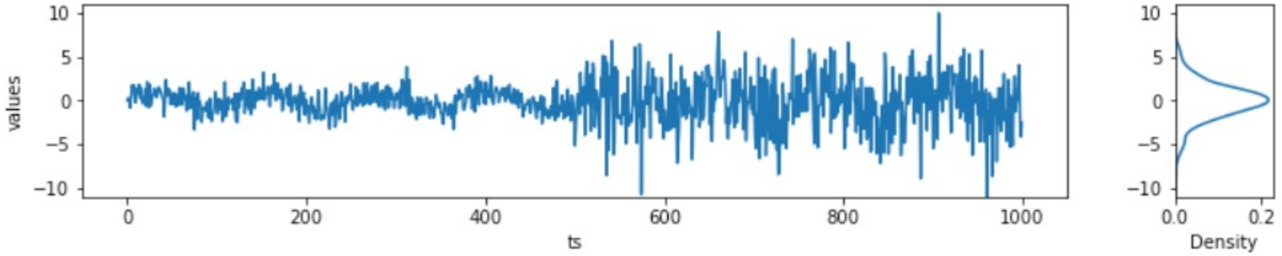
No repetitive pattern (a.k.a. **seasonality**)



Stationarity
No combinations of the previous ones :-P

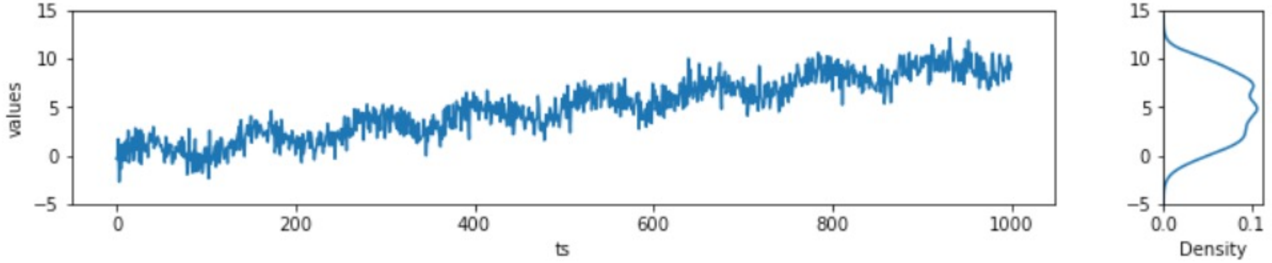
non-constant variance

+ seasonality



non-constant mean

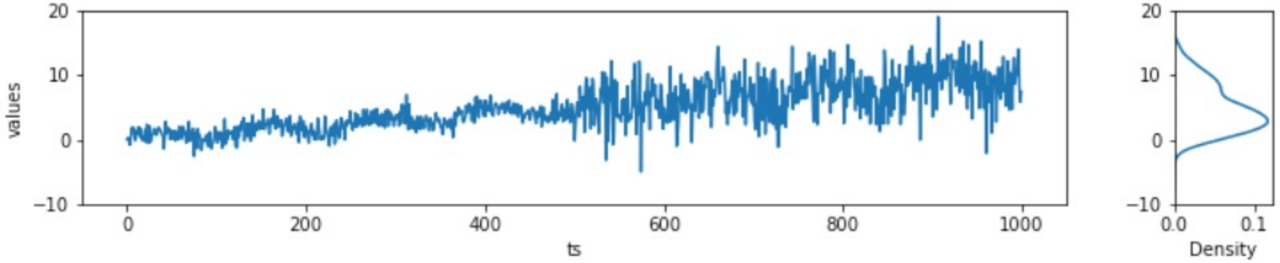
+ seasonality



Non-constant mean

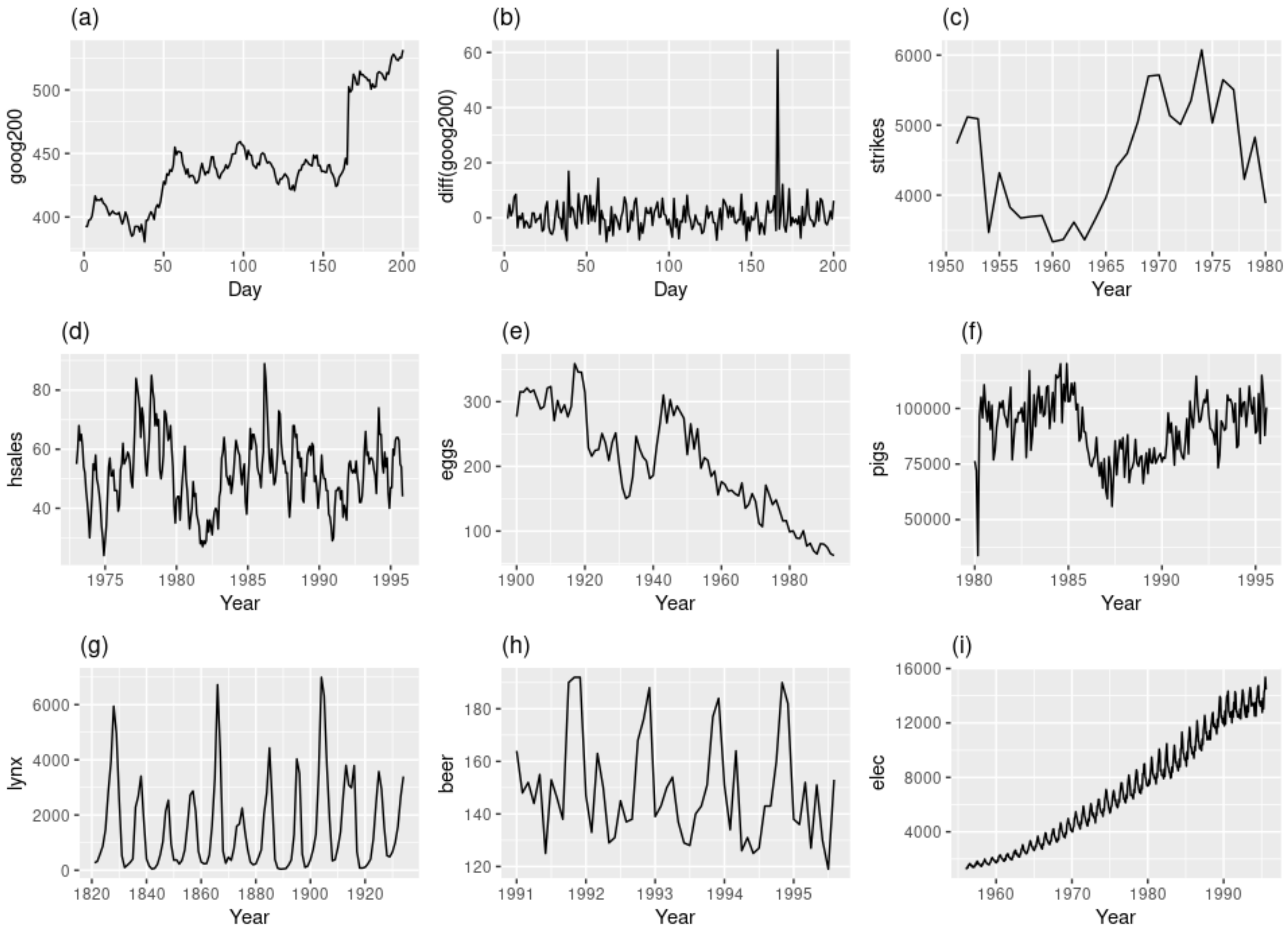
+ non-constant variance

+ seasonality



Stationarity
Let’s see if you got the point

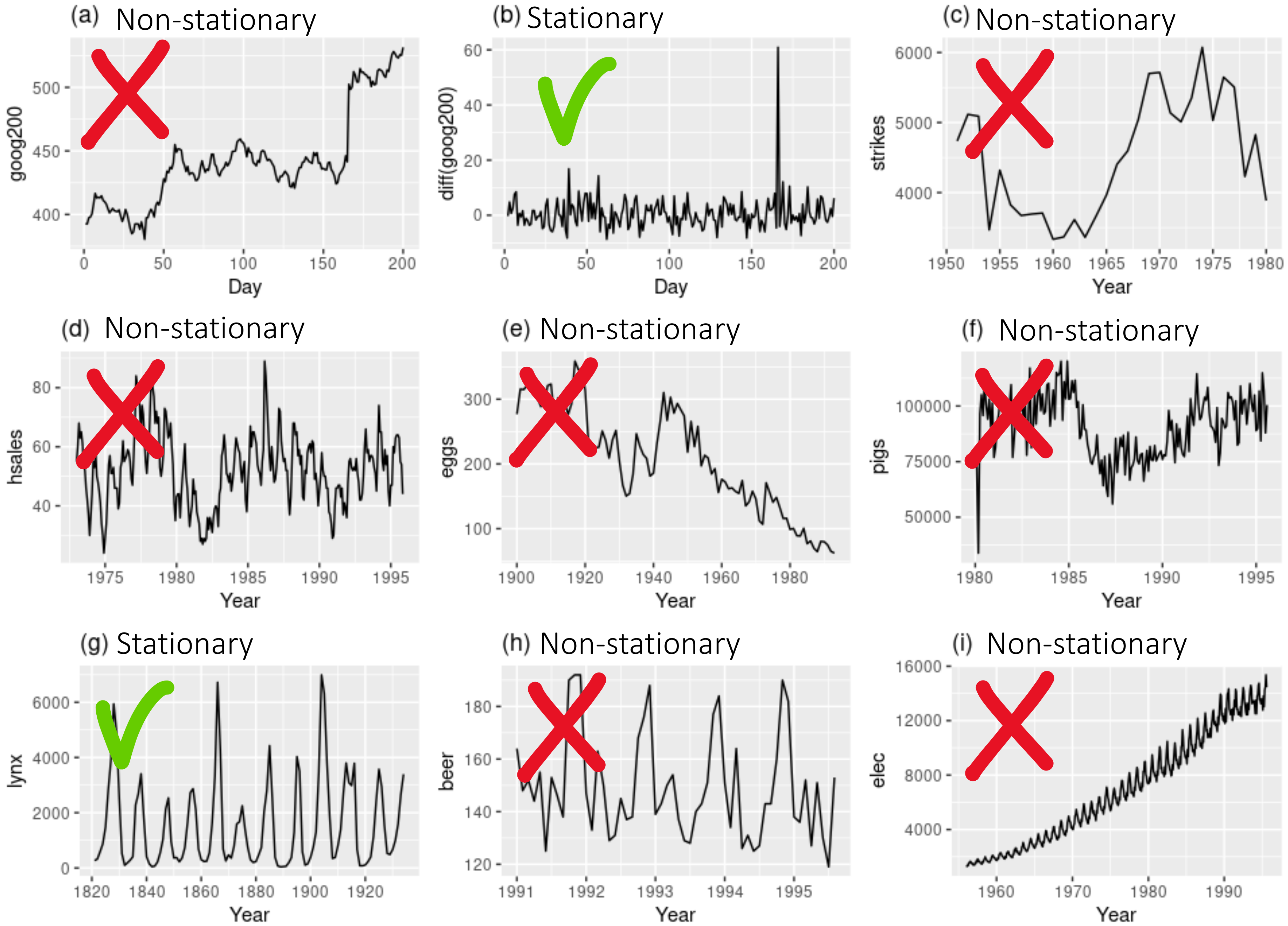
- Which of these time series is stationary?
- Why?



Quiz

stationary

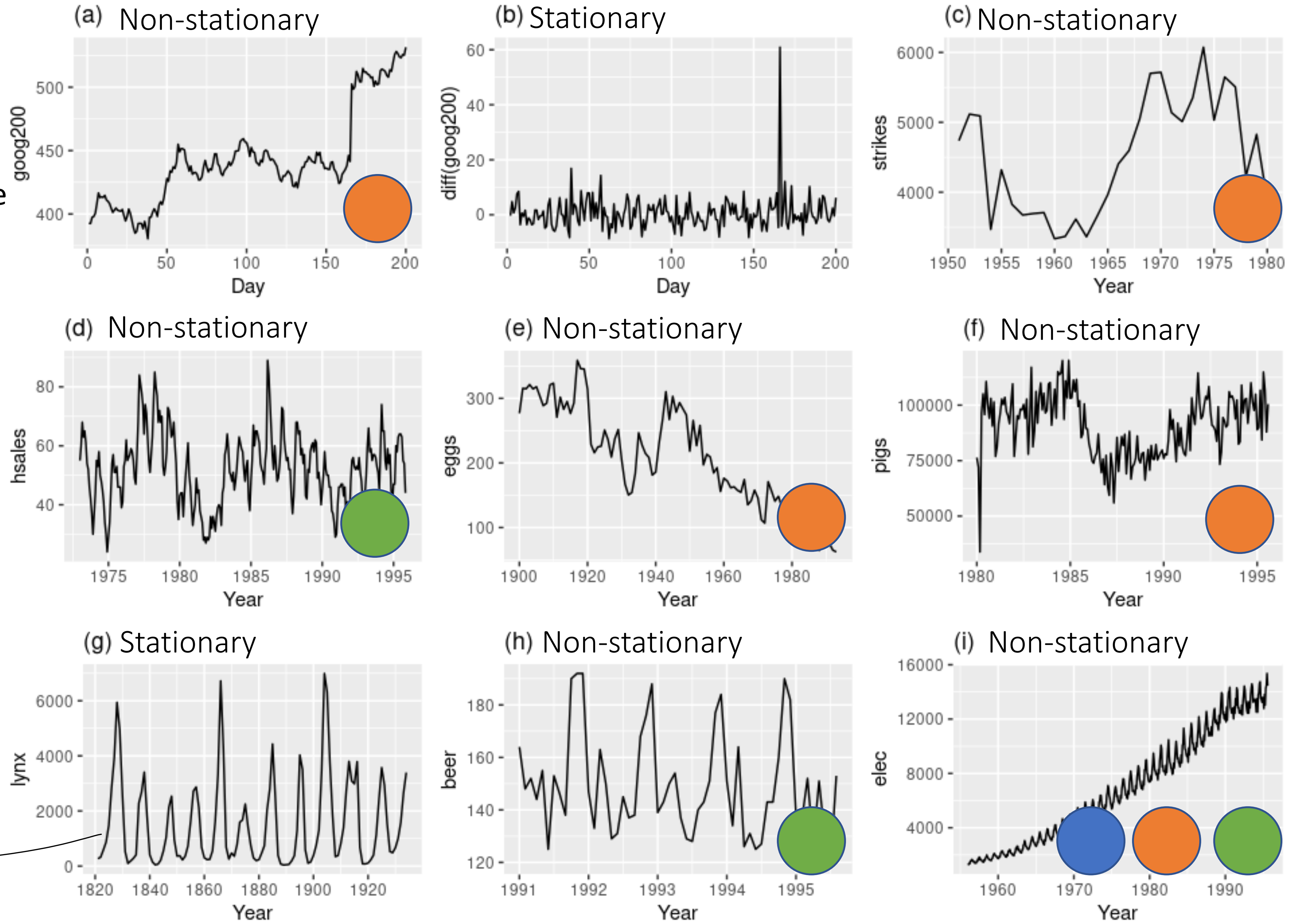
non-stationary



Non-constant mean

Non-constant variance

With seasonality



Irregular cycles
At first glance, the strong cycles might appear to make it non-stationary. But these cycles are aperiodic — they are caused when the lynx population becomes too large for the available feed, so that they stop breeding and the population falls to low numbers, then the regeneration of their food sources allows the population to grow again, and so on. In the long-term, the timing of these cycles is not predictable. Hence the series is stationary.

Formally…

Photo by **George Becker** from **Pexels**

Stationary

Formal definition

- Let $\{ X_t \}$ be a stochastic process and let $F_X(t_{1+\tau}, \dots, t_{k+\tau})$ represent the cumulative distribution function of the unconditional (i.e., with no reference to any particular starting value) joint distribution of $\{ X_t \}$ at times $t_{1+\tau}, \dots, t_{k+\tau}$.
- Then, $\{ X_t \}$ is said to be **strictly stationary, strongly stationary or strict-sense stationary** if

$$F_X(t_1, \dots, t_k) = F_X(t_{1+\tau}, \dots, t_{k+\tau}) \text{ for any } \tau \; e \; k.$$

How to **test** for **stationarity**

Stationary

By hand …

1. Load a time series
2. Split it two parts
3. Compute mean and variance of the two parts
4. Compare them

Statistical tests for stationarity

- Beyond simple checks on mean and variance, we can rely on **formal statistical tests** to assess stationarity.
- The most common are the **Augmented Dickey-Fuller (ADF)** and **Kwiatkowski-Phillips-Schmidt-Shin (KPSS)** tests, providing a more robust evaluation of whether a time series is stationary or not.

Unit root tests

When analyzing time series stationarity, the key question is whether the **data follow a stable pattern or drift over time**. This behavior is captured by the idea of a unit root.

A unit root means the current value of the time series depends strongly on its previous value, plus some random noise:

$$y_t = \phi y_{t-1} + \varepsilon_t$$

When $\phi = 1$, shocks accumulate rather than dissipate, causing the series to wander instead of reverting to a long-term mean.

In practice, the presence of a **unit root indicates non-stationarity**.

- ADF test: checks whether a unit root is present
- KPSS test: checks whether a unit root is absent (i.e., the series is stationary)

Augmented Dickey-Fuller (ADF) Test

The ADF test is one of the most common tools to check for unit roots, and therefore for non-stationarity. It tests whether the time series can be described as a random walk (non-stationary) or as a mean-reverting process (stationary).

- Null hypothesis (H_0): the series has a unit root → non-stationary
- Alternative hypothesis (H_1): the series is stationary

In practice:

- If the **p-value is low** (usually < 0.05) → reject H_0 → the **series is stationary**
- If the **p-value is high** (usually ≥ 0.05) → fail to reject H_0 → the **series is non-stationary**

Kwiatkowski-Phillips-Schmidt-Shin (KPSS)

The KPSS test is another widely used method to assess stationarity, but it takes the opposite perspective of the ADF test. Instead of testing for a unit root, it tests whether the series is stationary around a deterministic trend.

- Null hypothesis (H_0): the series is stationary
- Alternative hypothesis (H_1): the series is non-stationary

In practice:

- If the **p-value is low** (usually < 0.05) → reject H_0 → the **series is non-stationary**
- If the **p-value is high** (usually ≥ 0.05) → fail to reject H_0 → the **series is stationary**

Comparing ADF and KPSS - Two Views of Stationarity

Both ADF and KPSS tests check whether a **series is stationary**, but they rely on **different assumptions** about what “stationary” means.

	ADF test	**KPSS test**
Null hypothesis (H_0)	Series has a **unit root** (non-stationary)	Series is **stationary**
Alternative (H_1)	Series is **stationary**	Series is **non-stationary**
Underlying idea	**Difference-stationary** (stochastic trend)	**Trend-stationary** (deterministic trend)
Typical use	Detects presence of a unit root	Confirms stability around a trend
Interpretation	Low *p* → stationary	Low *p* → non-stationary

Stationary
Statistical tests

- Test for stationarity using statistical tests.
- ADF test: Augmented Dickey Fuller test
- KPSS test: Kwiatkowski-Phillips-Schmidt-Shin test

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